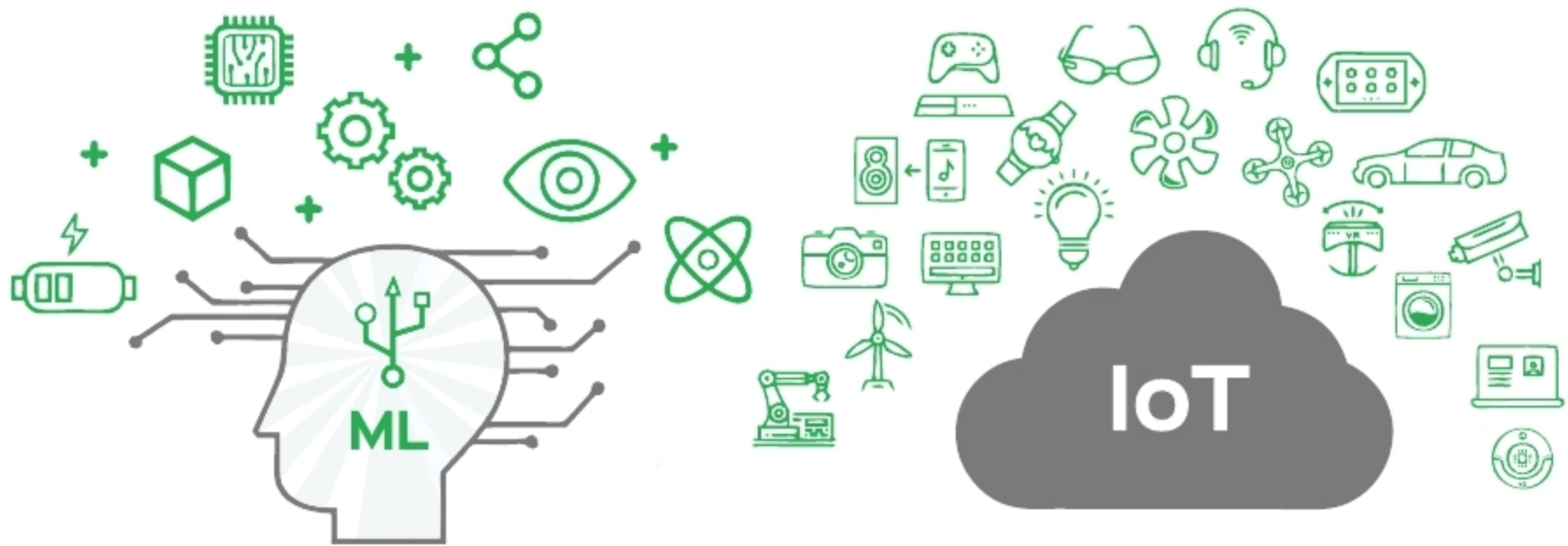


Machine Learning (ML) and IoT can Work Together to Improve Lives

IoT devices are becoming popular nowadays. The widespread use of IoT yields huge amounts of raw data. This data can be effectively processed by using machine learning to derive many useful insights that can become game changers and affect our lives deeply.



The field of machine learning is growing steadily, along with the growth of the IoT. Sensors, nano cameras, and other such IoT elements are now ubiquitous, placed in mobile phones, computers, parking stations, traffic control centres and even in home appliances. There are millions of IoT devices in the world and more are being manufactured every day. They collect huge amounts of data that is fed to machines via the Internet, enabling machines to ‘learn’ from the data and make them more efficient.

In IoT, it is important to note that a single device/element can generate immense amounts of data every second. All this data from IoT is transmitted to servers or gateways to create better machine learning models. Data analytics software can convert this raw data into useful insights so that the machine can be made more intelligent, and perform better with cost-effectiveness and a long life. By the year 2020, the world will have an estimated 20 billion IoT devices. Data collected by these devices mostly pertains to machines. By using this data, machines can learn more effectively and

can overcome their own drawbacks.

Now let's look at how machine learning and IoT can be combined together. Let us suppose that I have some bananas and apples. I have got a sophisticated nano camera and sensors to collect the data from these fruits. If the data collected by these elements is fed to my laptop through the Internet, my laptop will start analysing the information by using sophisticated data analytics software and the cloud platform. Now if my laptop shows graphically how many bananas and apples I have left, it probably means that my machine (laptop) hasn't learnt enough. On the other hand, if my laptop is able to describe graphically how many of these are now ripe enough to be eaten, how many are not quite ripe and how many are very raw, it proves that my machine (laptop) has learned enough and has become more intelligent.

Storing, processing, analysing and being able to ‘reason out’ using IoT data requires numerous computational and financial resources to attain business and machine learning values.

Today an Airbus aircraft is provided with thousands of sensors to measure temperature, speed, fuel consumption, air flow dynamics, mechanisms of working, etc. All this data provided by IoT devices is connected to cloud platforms such as IBM Watson, Microsoft Azure, etc, via the Internet. Using sophisticated data analytics software, useful information is fed back to the machine, i.e., the aircraft. Using this data, the machine can learn very fast to overcome its problems, so that its life span and performance can be greatly enhanced.

Today, the IoT connects several sectors such as manufacturing industries, healthcare, buildings, vehicles, traffic, shopping centres and so on. Data gathered from such diverse domains can certainly make the infrastructure learn meaningfully to work more efficiently.

Giving a new deal to electronic vision

Amazon DeepLens is a wireless-enabled video camera and is integrated with Amazon Cloud. It makes use of